

Exporting An Employee's Microsoft 365 Data to Your Computer

What this does

This guide lets you extract a specific employee's Microsoft 365 data — emails, OneDrive files, and Teams chats — to your computer, for a date range you choose. It uses a Python script that talks directly to Microsoft's API.

Data type	What you get	Format
Outlook emails	All emails + attachments in the date range	JSON + files
Teams 1:1 chats	Personal chats (stored inside mailbox)	JSON
OneDrive files	Docs, PPTs, Excel, PDFs — original folder structure	Native files
Teams channels	Channel messages (needs one extra permission)	JSON
Summary sheet	emails_metadata.csv — open in Excel	CSV

Before you start

Microsoft 365 plan	Business Basic or Standard (or any higher plan)
Your role	Global Administrator on the Microsoft 365 tenant
Python	Version 3.10 or newer — download from python.org
Computer	Mac or Windows — the script runs in Terminal / PowerShell
Time needed	~20 min setup (one-time) + script runs in the background

Overview — 3 stages

The whole process splits into 3 stages. Stages 1 and 2 happen once. Stage 3 is what you run each time you want to export someone's data.

Stage 1: Register an app in Microsoft Azure (*in portal.azure.com*)

Stage 2: Add credentials to the script (*in .env file*)

Stage 3: Run the export script (*in terminal*)

Stage 1 — Register an app in Microsoft Azure (in portal.azure.com)

Think of this like creating a service account for your script — Microsoft needs to know who is making requests before it hands over any data. You do this in the Azure portal.

Step 1 — Sign in and create the app

1. Go to **portal.azure.com** → sign in with your admin account
2. Search **Microsoft Entra ID** in the top search bar → click it
3. Left sidebar: **App registrations** → **+ New registration**
4. Name it **m365-exporter** → leave everything else as default → click **Register**

Step 2 — Copy your 3 credentials

On the app's **Overview page**, copy and save these two values somewhere safe:

5. **Application (client) ID** — a long string like a1b2c3d4-...
6. **Directory (tenant) ID** — another long string below it

Then **create your third credential (the password)**:

7. Left sidebar: **Certificates & secrets** → **+ New client secret**
8. Add a description like **exporter-key** → set expiry to **1 month** → click **Add**
9. **Copy the Value immediately** — it only shows once. This is your **CLIENT_SECRET**.

⚠ If you navigate away without copying the secret, you cannot retrieve it. You would have to create a new one.

Step 3 — Add permissions

This controls what the script is allowed to read.

10. Left sidebar: **API permissions** → **+ Add a permission** → **Microsoft Graph** → **Application permissions**
11. Search and add: **Mail.Read**
12. Search and add: **Files.Read.All**
13. Search and add: **User.Read.All**
14. Click **Grant admin consent for [your org]** at the top → confirm **Yes**
15. All 3 permissions should now show a **green tick** under Status

✓ The green tick confirms Microsoft has authorised the script. Without it, every API call will return a 403 Forbidden error.

Stage 2 — Set up the script on your computer (in Terminal)

Step 1 — Install Python

16. Go to **python.org** → download Python 3.10 or newer
17. On Windows: tick "**Add Python to PATH**" during install — important
18. Open Terminal (Mac) or PowerShell (Windows) and check it worked:

```
python3 --version
```

You should see a number like 3.11.x. (On Windows you may need to type `python --version.`)

Step 2 — Download and set up the script

19. Open Terminal and clone the repo from GitHub:

```
git clone https://github.com/anujchahar-lh2/m365-exporter.git
```

20. Navigate into the folder:

```
cd m365-exporter
```

21. Install the script's dependencies (one-time):

```
python3 -m pip install -r requirements.txt
```

Step 3 — Add your credentials

22. Copy the example settings file:

```
cp .env.example .env
```

23. Open the `.env` file in any text editor and fill in your 3 values from Stage 1:

```
TENANT_ID=paste-your-directory-tenant-id-here  
CLIENT_ID=paste-your-application-client-id-here  
CLIENT_SECRET=paste-your-client-secret-value-here
```

⚠ Never share this `.env` file or commit it to Git. It contains the keys to your Microsoft 365 tenant.

Stage 3 — Run the export (in Terminal)

Copy and paste one of these commands into your terminal. Replace the email addresses with the real ones.

Export one employee — all time

```
python3 run_m365_export.py \  
  --admin admin@yourcompany.com \  
  --user employee@yourcompany.com \  
  --export-only --local-only
```

Export one employee — last 90 days only

```
python3 run_m365_export.py \  
  --admin admin@yourcompany.com \  
  --user employee@yourcompany.com \  
  --export-only --local-only \  
  --since-days 90
```

Export one employee — from a specific date

```
python3 run_m365_export.py \  
  --admin admin@yourcompany.com \  
  --user employee@yourcompany.com \  
  --export-only --local-only \  
  --modified-after 2024-01-01
```

Export two or more employees at once

```
python3 run_m365_export.py \  
  --admin admin@yourcompany.com \  
  --only alice@yourcompany.com \  
  --only bob@yourcompany.com \  
  --export-only --local-only \  
  --since-days 90
```

✓ The script can run for hours on large mailboxes. You can leave it going and come back. If it stops, run the same command again — it resumes where it left off automatically.

Common issues

Problem	Fix
401 Unauthorized	Check TENANT_ID, CLIENT_ID, CLIENT_SECRET in your .env file — one of them is wrong or has extra spaces.
403 Forbidden	Admin consent wasn't granted. Go back to Stage 1 Step 3 and click 'Grant admin consent'.
python3: command not found	Python isn't installed or not on PATH. Reinstall from python.org. On Windows, tick 'Add to PATH'.
ModuleNotFoundError	Run: <code>python3 -m pip install -r requirements.txt</code>
Script stopped midway	Run the same command again. It skips already-downloaded files and continues from where it stopped.
Employee not found (404)	Double-check the email address spelling. The account must be active in Microsoft 365.
out/ folder is empty	The date range may be too narrow. Try removing <code>--since-days</code> or <code>--modified-after</code> to get everything.
CLIENT_SECRET expired	Secrets expire after the period you set (e.g. 12 months). Create a new one in Azure → Certificates & secrets → New client secret.

Clean up after use: Once the export is handed off, delete the out/ folder and revoke the client secret in Azure (portal.azure.com → your app → Certificates & secrets → delete).